

```
1 coins = [1, 2, 5, 10]
2 amount = 28
3
4 coins.sort(reverse=True)
5
6 count = 0
7
8 for coin in coins:
9     while amount >= coin:
10         print(coin, end=" ")
11         amount = amount - coin
12         count = count + 1
13
14 print()
15 print("Minimum coins =", count)
```



```
10 10 5 2 1
Minimum coins = 5
```

```
1 coins = [1, 5, 10, 25]
2 amount = 63
3
4 coins.sort(reverse=True)
5
6 count = 0
7
8 for coin in coins:
9     while amount >= coin:
10         print(coin, end=" ")
11         amount = amount - coin
12         count = count + 1
13
14 print()
15 print("Minimum coins =", count)
```

✓ ↗ 📄 ⚙️ 📌

25 25 10 1 1 1

Minimum coins = 6

```
1 coins = [1, 3, 7, 10]
2 amount = 24
3
4 coins.sort(reverse=True)
5
6 count = 0
7
8 for coin in coins:
9     while amount >= coin:
10         print(coin, end=" ")
11         amount = amount - coin
12         count = count + 1
13
14 print()
15 print("Minimum coins =", count)
```

10 10 3 1  
Minimum coins = 4

```
1 weights = [10, 20, 30]
2 profits = [60, 100, 120]
3 capacity = 50
4
5 items = []
6
7 for i in range(len(weights)):
8     ratio = profits[i] / weights[i]
9     items.append((ratio, weights[i], profits[i]))
10
11 items.sort(reverse=True)
12
13 total = 0
14
15 for ratio, weight, profit in items:
16     if capacity >= weight:
17         capacity = capacity - weight
18         total = total + profit
19     else:
20         total = total + ratio * capacity
21         capacity = 0
22         break
23
24 print("Maximum profit =", total)
```



Maximum profit = 240.0

```
1 weights = [5, 10, 15]
2 profits = [30, 40, 45]
3 capacity = 20
4
5 items = []
6
7 for i in range(len(weights)):
8     ratio = profits[i] / weights[i]
9     items.append((ratio, weights[i], profits[i]))
10
11 items.sort(reverse=True)
12
13 total = 0
14
15 for ratio, weight, profit in items:
16     if capacity >= weight:
17         capacity = capacity - weight
18         total = total + profit
19     else:
20         total = total + ratio * capacity
21         capacity = 0
22         break
23
24 print("Maximum profit =", total)
```



Maximum profit = 85.0



```
1 weights = [4, 8, 2, 6]
2 profits = [20, 40, 14, 30]
3 capacity = 10
4
5 items = []
6
7 for i in range(len(weights)):
8     ratio = profits[i] / weights[i]
9     items.append((ratio, weights[i], profits[i]))
10
11 items.sort(reverse=True)
12
13 total = 0
14
15 for ratio, weight, profit in items:
16     if capacity >= weight:
17         capacity = capacity - weight
18         total = total + profit
19     else:
20         total = total + ratio * capacity
21         capacity = 0
22         break
23
24 print("Maximum profit =", total)
```



Maximum profit = 54.0

```

2 deadlines = [2, 1, 2, 1]
3 profits = [100, 19, 27, 25]
4
5 data = []
6
7 for i in range(len(jobs)):
8     data.append((profits[i], deadlines[i], jobs[i]))
9
10 data.sort(reverse=True)
11
12 slots = [None, None]
13 total = 0
14
15 for profit, deadline, job in data:
16     for j in range(deadline - 1, -1, -1):
17         if slots[j] is None:
18             slots[j] = job
19             total = total + profit
20             break
21
22 print("Selected jobs =", end=" ")
23
24 for job in slots:
25     if job is not None:
26         print(job, end=" ")
27
28 print()
29 print("Maximum profit =", total)

```



Selected jobs = C A

Maximum profit = 127

```

2 deadlines = [2, 1, 2, 1, 3]
3 profits = [20, 15, 10, 5, 1]
4
5 data = []
6
7 for i in range(len(jobs)):
8     data.append((profits[i], deadlines[i], jobs[i]))
9
10 data.sort(reverse=True)
11
12 slots = [None, None, None]
13 total = 0
14
15 for profit, deadline, job in data:
16     for j in range(deadline - 1, -1, -1):
17         if slots[j] is None:
18             slots[j] = job
19             total = total + profit
20             break
21
22 print("Selected jobs =", end=" ")
23
24 for job in slots:
25     if job is not None:
26         print(job, end=" ")
27
28 print()
29 print("Maximum profit =", total)

```



Selected jobs = J2 J1 J5  
Maximum profit = 36



```

1 deadlines = [2, 1, 2, 1, 3]
2 profits = [100, 50, 10, 20, 30]
3
4 data = []
5
6
7 for i in range(len(jobs)):
8     data.append((profits[i], deadlines[i], jobs[i]))
9
10 data.sort(reverse=True)
11
12 slots = [None, None, None]
13 total = 0
14
15 for profit, deadline, job in data:
16     for j in range(deadline - 1, -1, -1):
17         if slots[j] is None:
18             slots[j] = job
19             total = total + profit
20             break
21
22 print("Selected jobs =", end=" ")
23
24 for job in slots:
25     if job is not None:
26         print(job, end=" ")
27
28 print()
29 print("Maximum profit =", total)

```



Selected jobs = B A E  
Maximum profit = 180

```

2 deadlines = [1, 1, 2, 2]
3 profits = [40, 30, 20, 10]
4
5 data = []
6
7 for i in range(len(jobs)):
8     data.append((profits[i], deadlines[i], jobs[i]))
9
10 data.sort(reverse=True)
11
12 slots = [None, None]
13 total = 0
14
15 for profit, deadline, job in data:
16     for j in range(deadline - 1, -1, -1):
17         if slots[j] is None:
18             slots[j] = job
19             total = total + profit
20             break
21
22 print("Selected jobs =", end=" ")
23
24 for job in slots:
25     if job is not None:
26         print(job, end=" ")
27
28 print()
29 print("Maximum profit =", total)

```



Selected jobs = P1 P3  
Maximum profit = 60

```

4 nodes = []
5
6 for i in range(len(chars)):
7     nodes.append([freq[i], chars[i]])
8
9 codes = {}
10
11 for ch in chars:
12     codes[ch] = ""
13
14 while len(nodes) > 1:
15     nodes.sort()
16
17     left = nodes.pop(0)
18     right = nodes.pop(0)
19
20     for ch in left[1]:
21         codes[ch] = "0" + codes[ch]
22
23     for ch in right[1]:
24         codes[ch] = "1" + codes[ch]
25
26     nodes.append([left[0] + right[0], left[1] + right[1]])
27
28 print("Huffman Codes:")
29
30 for ch in chars:
31     print(ch + ":", codes[ch])

```



```

c: 100
d: 101
e: 111
f: 0

```

```

4 nodes = []
5
6 for i in range(len(chars)):
7     nodes.append([freq[i], chars[i]])
8
9 codes = {}
10
11 for ch in chars:
12     codes[ch] = ""
13
14 while len(nodes) > 1:
15     nodes.sort()
16
17     left = nodes.pop(0)
18     right = nodes.pop(0)
19
20     for ch in left[1]:
21         codes[ch] = "0" + codes[ch]
22
23     for ch in right[1]:
24         codes[ch] = "1" + codes[ch]
25
26     nodes.append([left[0] + right[0], left[1] + right[1]])
27
28 print("Huffman Codes:")
29
30 for ch in chars:
31     print(ch + ":", codes[ch])

```



```

A: 100
B: 101
C: 11
D: 0

```



```

4 nodes = []
5
6 for i in range(len(chars)):
7     nodes.append([freq[i], chars[i]])
8
9 codes = {}
10
11 for ch in chars:
12     codes[ch] = ""
13
14 while len(nodes) > 1:
15     nodes.sort()
16
17     left = nodes.pop(0)
18     right = nodes.pop(0)
19
20     for ch in left[1]:
21         codes[ch] = "0" + codes[ch]
22
23     for ch in right[1]:
24         codes[ch] = "1" + codes[ch]
25
26     nodes.append([left[0] + right[0], left[1] + right[1]])
27
28 print("Huffman Codes:")
29
30 for ch in chars:
31     print(ch + ":", codes[ch])

```



```

x: 110
y: 111
z: 10
w: 0

```

```
1 coins = [1, 2, 5]
2 amount = 11
3
4 coins.sort(reverse=True)
5
6 count = 0
7
8 for coin in coins:
9     while amount >= coin:
10         print(coin, end=" ")
11         amount = amount - coin
12         count = count + 1
13
14 print()
15 print("Minimum coins =", count)
```



5 5 1

Minimum coins = 3

```
1 weights = [2, 3, 5, 7]
2 profits = [10, 5, 15, 7]
3 capacity = 8
4
5 items = []
6
7 for i in range(len(weights)):
8     ratio = profits[i] / weights[i]
9     items.append((ratio, weights[i], profits[i]))
10
11 items.sort(reverse=True)
12
13 total = 0
14
15 for ratio, weight, profit in items:
16     if capacity >= weight:
17         capacity = capacity - weight
18         total = total + profit
19     else:
20         total = total + ratio * capacity
21         capacity = 0
22         break
23
24 print("Maximum profit =", total)
```



Maximum profit = 26.666666666666668